

Rahul Gomes, Ph.D.

Associate Professor & Chair, Department of Computer Science
University of Wisconsin–Eau Claire

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ACADEMIC APPOINTMENTS

Associate Professor of Computer Science , University of Wisconsin–Eau Claire	Aug 2024–Present
Research Temporary Professor , Mayo Clinic	2026–Present
AI & Informatics (AI&I); collaboration on undergraduate research internships and applied healthcare artificial intelligence.	
Ramsey Research Professor , University of Wisconsin–Eau Claire	2025–2028
Assistant Professor of Computer Science , University of Wisconsin–Eau Claire	Jul 2020–Jul 2024
Assistant Professor of Computer Science , Minot State University	Aug 2019–Jun 2020
Instructor of Computer Science , Minot State University	Aug 2018–Aug 2019
Adjunct Lecturer , Dakota College at Bottineau	Aug 2019–Dec 2019
Computer Science Instructor , Johns Hopkins Center for Talented Youth	Jul 2018–Aug 2018

ACADEMIC AND INSTITUTIONAL LEADERSHIP

- **Chair, Department of Computer Science, University of Wisconsin–Eau Claire.** Provide academic and administrative leadership for the department, including curriculum, personnel, enrollment, program development, external partnerships, and strategic initiatives. 2026–Present
- **Faculty Coordinator, Blugold Center for High Performance Computing.** Contribute to the development and use of shared high-performance computing, GPU, and AI research infrastructure supporting interdisciplinary research, education, and student training. 2026–Present
- **Artificial Intelligence Program Development.** Played a significant collaborative role in developing UW–Eau Claire’s undergraduate artificial intelligence programs launched in Fall 2025, including curriculum and course development, stakeholder engagement, and efforts to establish external partnerships and an AI advisory board.
- **Campus Artificial Intelligence Strategy.** Participate in university-level AI planning discussions with university leadership, faculty, and staff addressing instructional and operational AI resources, research and computing infrastructure, institutional policy, and student outcomes. 2026–Present
- **Campus AI Education Initiative.** Working with university administration and the Center for Excellence in Teaching and Learning (CETL) to develop recurring campus discussions for faculty and staff addressing emerging AI technologies, practices, opportunities, and challenges. 2026–Present
- **Universities of Wisconsin AI Engagement.** Contributed to broader system-level conversations on artificial intelligence, including participation in an AI panel before the Universities of Wisconsin Board of Regents.
- **Faculty Administrator, Blugold Center for High Performance Computing.** Supported interdisciplinary use of high-performance computing infrastructure, undergraduate research, faculty research, and research-computing education. 2020–2024

EDUCATION

North Dakota State University <i>Ph.D., Computer Science</i>	Fargo, ND Dec 2019
Sikkim Manipal University <i>M.S., Computer Science</i>	Sikkim, India Apr 2014
St. Xavier's College <i>B.Ed.</i>	Kolkata, India Aug 2012
St. Xavier's College <i>B.S.</i>	Kolkata, India Jun 2010

RESEARCH INTERESTS

Artificial intelligence; machine learning; deep learning; high-performance computing; medical imaging and healthcare AI; cancer informatics and multi-omics; geospatial artificial intelligence; precision agriculture; scientific machine learning; research cyberinfrastructure; and AI-enabled education and simulation.

HONORS AND AWARDS

- **Mid-Career Faculty Mentor Award**, Mathematical, Computing, and Statistical Sciences Division, Council on Undergraduate Research (CUR). 2026
- **Ramsey Research Professor**, University of Wisconsin–Eau Claire and Mayo Clinic Health System. 2025–2028
- **Top 10 Abstract**, Mayo Clinic AI Summit, Rochester, Minnesota. 2025
- **Collaborator of the Year**, Mayo Clinic Health System, Northwest Wisconsin. 2024
- **Emerging Mentor in Research, Scholarship and Creative Activity**, University of Wisconsin–Eau Claire. 2024
- **Best Oral Presentation Award**, International Conference on Spatial Statistics and Geostatistics, New York, NY. 2019
- **Academic Excellence in Computer Science Award**, North Dakota State University College of Science and Mathematics. 2018
- **Rahul Devabhaktuni Memorial Scholarship**, North Dakota State University Department of Computer Science. 2017, 2018
- **Third Place, Best Speaker**, Midwest Instruction and Computing Symposium. 2018
- **First Place, Best Speaker**, NDSU Graduate Research Symposium. 2017

SPONSORED RESEARCH AND GRANTS

Active External Awards

- National Science Foundation – Research Experiences for Undergraduates** 2025–2028
REU Site: Advancing High-Performance Computing Opportunities in Undergraduate Research at UW–Eau Claire to Meet Challenges of Multidisciplinary Computational Science. NSF OAC-2447779. \$459,810. Principal Investigator: **Rahul Gomes**; Co-PI: Ying Ma.
- National Science Foundation – Research Experiences for Teachers** 2024–2027
Teachers As Researchers in Computing Classrooms (TARCC): Bridging Gaps in High School Computing Education in Western Wisconsin and Eastern Minnesota. NSF CNS-2419269. \$550,786. PI: Anthony Varghese; Co-PIs: **Rahul Gomes**, Katrina Rothrock, and Abhimanyu Ghosh.

Completed External and Collaborative Awards

- National Science Foundation – Research Experiences for Undergraduates** 2022–2026
REU Site: Advancing High-Performance Computing Opportunities in Undergraduate Research at UW–Eau Claire to Meet Challenges of Multidisciplinary Computational Science. NSF OAC-2150191. \$384,762. Principal Investigator: **Rahul Gomes**; Co-PI: Sudeep Bhattacharyya.
- UW–Eau Claire–Mayo Clinic Research Innovation Council** 2023–2025
Retrospective Institutional Comparison of Patients Undergoing “On-Pump” Versus “Off-Pump” Coronary Artery Bypass Grafting Using Machine Learning. \$9,999.44. PI: Nishant Saran; Co-PI: **Rahul Gomes**.
- Tommy G. Thompson Center on Public Leadership** 2021–2022
Exploring Policies to Promote High-Performance Computing in Post-Pandemic Undergraduate Education in Wisconsin. \$94,368. PI: Ying Ma; Co-PIs: **Rahul Gomes** and Sudeep Bhattacharyya.
- NDSU Center for Diagnostic & Therapeutic Strategies in Pancreatic Cancer** 2020–2022
Using Methylation and Gene Expression Data for Early Detection of Pancreatic Cancer. \$15,000. PI: Rick Jansen; Co-PI: **Rahul Gomes**.
- UW–Eau Claire–Mayo Clinic Research Innovation Council** 2021–2022
Detection of Inferior Vena Cava Filters on CT Scans Using Artificial Intelligence. \$33,965. PI: Joe Wildenberg; Co-PI: **Rahul Gomes**.
- UW–Eau Claire–Mayo Clinic Research Innovation Council** 2021–2023
Predicting Chemotherapy Outcomes in Pancreatic Ductal Adenocarcinoma Patients Using CT Imaging as a Biomarker with Artificial Intelligence. Mayo PI: Sushil Kumar Garg; UWEC PI: **Rahul Gomes**.
- Biomedical Innovation Grant – Mayo Collaborative Research Program** 2021–2023
Comparison of Pathways in Humans and Genetically Engineered Mouse Models for Pancreatic Ductal Adenocarcinoma. \$15,000. NDSU PI: Rick Jansen; UWEC PI: **Rahul Gomes**.

Selected Internal and Seed Funding

- **Deep Learning Segmentation of Kidney Tissue Microarrays Using Infrared Spectral Imaging.** UW–Eau Claire ORSP. PI: **Rahul Gomes**; Co-PI: Michael J. Walsh. \$5,300. 2021–2022
- **Exploring an Optimized Deep Learning Framework for Analysis of Satellite Imagery.** UW–Eau Claire ORSP. PI: **Rahul Gomes**; Co-PI: Papia Rozario. \$13,200. 2020–2022
- **A Hybrid Deep Learning Model for Prediction of Severity in COVID-19 Patients.** UW–Eau Claire ORSP. PI: **Rahul Gomes**. \$2,500. 2021–2022
- **NDCyberGIS: Integrating Big Data with Intelligent Knowledge Discovery NoSQL Systems for Geovisualization and Machine Learning.** ND EPSCoR Seed Award. PI: **Rahul Gomes**; Co-PI: Anne Denton. \$9,986.24. 2019–2020
- **CyberPatriot Outreach Program.** ND EPSCoR Seed Award. PI: Darren Seifert; Co-PIs: **Rahul Gomes** and Sayeed Sajal. \$5,987.34. 2019–2020

UNDERGRADUATE RESEARCH AND MENTORING

- Mentored more than **70 undergraduate researchers** across computer science, healthcare, bioinformatics, geospatial science, and high-performance computing. Student collaborations have resulted in at least **17 peer-reviewed publications**, 34 institutional and regional conference presentations, and 23 CERCA presentations involving 53 student presenters.
- **Principal Investigator, NSF REU Site.** Direct a multidisciplinary summer undergraduate research program centered on high-performance computing and computational science. The program supports approximately ten undergraduate researchers each summer and integrates faculty mentorship, research computing, professional development, and dissemination.

- **Mayo Clinic AI & Informatics Undergraduate Research.** Develop and mentor undergraduate research and internship experiences with Mayo Clinic collaborators in healthcare AI, including medical imaging, clinical text processing, patient education, colonoscopy and pathology report analysis, and retrieval-augmented generation.
- **NSF RET/TARCC.** Co-lead a research experience connecting computing research, research cyberinfrastructure, and secondary computing education for teachers from Western Wisconsin and Eastern Minnesota.

PEER-REVIEWED PUBLICATIONS

Underlined names indicate undergraduate student researchers.

1. Garg, Sushil Kumar, Mau, Brayden, Hubers, Jeffery, Arce, Victor, Hooper, Nicole, Lindquist, Jennifer, Harper, Sarah B., Thayer, Ann B., Mukherjee, Piyush, Loufek, Brenna T., Melnick, Joe, Rost, Lauren M., Robbins, Steven A., **Gomes, Rahul**, and Chaudhary, Rajeev. "Colon-Pilot: A Generative AI Tool for Automated Colonoscopy Surveillance Recommendations and 2024 ACG/ASGE Quality Benchmarking." *American Journal of Gastroenterology*, 121(4):964–973, 2026. <https://doi.org/10.14309/ajg.0000000000003946>
2. Mau, Brayden, Garg, Sushil Kumar, Harper, Sarah B., **Gomes, Rahul**, Rolli, Alexandra, and Chaudhry, Rajeev. "Distilling Knowledge in GI: An AI System for Assisting Colonoscopy and Pathology Report Review." *Mayo Clinic Proceedings: Digital Health*, 4(3):100372, 2026. <https://doi.org/10.1016/j.mcpdig.2026.100372>
3. Adeleke, David, Fadaka, Adewale Oluwaseun, Sibuyi, Nicole Remaliah Samantha, Klein, Ashwil, Meyer, Mervin, **Gomes, Rahul**, and Jansen, Rick. "AlphaMissense Pathogenicity Scores Predict Response to Immunotherapy and Enhances the Predictive Capability of Tumor Mutation Burden." *Translational Oncology*, 65:102697, 2026. <https://doi.org/10.1016/j.tranon.2026.102697>
4. Rozario, Papia F., Sipos, Gabriel, Dierkes, Landon, Tapia, Gabriel, Sanchez, Pedro, Sanyal, Debankur, Mitra, Dipankar, and **Gomes, Rahul**. "Integrating Soil Sensing and Machine Learning for Precision Agriculture: A Multiscale Approach to Smarter Fertility Management." *2026 IEEE International Conference on Electro Information Technology (eIT)*, pp. 1–7, 2026.
5. Saraboji, Kaaviyashri, **Gomes, Rahul**, and Mitra, Dipankar. "Lightweight Spatiotemporal Machine Learning Framework for Multi-Class Wildlife Roadkill Classification." *2026 IEEE International Conference on Electro Information Technology (eIT)*, pp. 319–323, 2026.
6. **Gomes, Rahul**, Jerome, William L., and Garg, Sushil K. "Integrating U-Net in a LLM Supervisor Agent Pipeline for Pancreatic Ductal Adenocarcinoma Diagnosis." *Mayo Clinic Proceedings: Digital Health*, 3(4), 2025. <https://doi.org/10.1016/j.mcpdig.2025.100287>
7. Buhrow, Mykle, Nie, Ian, Seliya, Naeem, Wildenberg, Joseph, and **Gomes, Rahul**. "YOLO-Based Object Detection for Automated Identification of IVC Filters in Abdominal CT Imaging." *2025 International Conference on Machine Learning and Applications (ICMLA)*, pp. 1318–1323, 2025.
8. Arunachalam, Valliammai, Rosen, Luke, Akinsiku, Mojisola Rachel, Dey, Shuvashis, **Gomes, Rahul**, and Mitra, Dipankar. "A Multi-Stage Deep Learning Framework for Antenna Array Synthesis in Satellite IoT Networks." *AI*, 6(10):248, 2025.
9. Arunachalam, Valliammai, Rosen, Luke, Mitra, Dipankar, Dey, Shuvashis, and **Gomes, Rahul**. "Efficient Prediction of Antenna Parameters Using Ensemble-Based Machine Learning Model." *2025 IEEE International Symposium on Antennas and Propagation and North American Radio Science Meeting (AP-S/CNC-USNC-URSI)*, pp. 1522–1525, 2025.
10. Rozario, Papia F., Chvat, Daniel, Dierkes, Landon, Abraham, Grace, North, Kristen, Sipos, Gabriel, McDonnell, Grace, and **Gomes, Rahul**. "Exploring Feature Selection and Cross-Validation for Improved Crop Yield Prediction." *2025 IEEE International Conference on Electro Information Technology (eIT)*, pp. 207–212, 2025.
11. Caterer, Zachary, Langlois, Jordan, McKeown, Connor, Hady, Mikayla, Stumo, Samuel, Setty, Suman, Walsh, Michael, and **Gomes, Rahul**. "Exploring Feature Selection with Deep Learning for Kidney Tissue Microarray Classification Using Infrared Spectral Imaging." *Bioengineering*, 12(4):366, 2025. <https://doi.org/10.3390/bioengineering12040366>

12. Wang, Andy, **Gomes, Rahul**, and Rozario, Papia F. “OSDA-ST: Advancing Open Set Domain Adaptation Through Selective Thresholding in Remote Sensing.” *2024 IEEE International Conference on Big Data (BigData)*, pp. 7563–7570, 2024. <https://doi.org/10.1109/BigData62323.2024.10825228>
13. Hebert, Jordan, Hratish, Ryan, **Gomes, Rahul**, Kunkel, William, Marshall, Daniel, Ghosh, Abhimanyu, Doss, Isabella, Ma, Ying, Stedman, Derek, Stinson, Blake, et al. “High-Performance Computing in Undergraduate Education at Primarily Undergraduate Institutions in Wisconsin: Progress, Challenges, and Opportunities.” *Education and Information Technologies*, 29(14):18451–18475, 2024. <https://doi.org/10.1007/s10639-024-12582-6>
14. Rozario, Papia F., Gadgil, Ravi, Lee, Junsu, **Gomes, Rahul**, Keller, Paige, Sipos, Gabriel, McDonnell, Grace, Impola, Westin, and Rudolph, Joseph. “Optimizing Mobile Vision Transformers for Land Cover Classification.” *Applied Sciences*, 14(13):5920, 2024. <https://doi.org/10.3390/app14135920>
15. Rozario, Papia F., Lee, Junsu, Chen, Yangguang, Mohan, Pavithra Devy, DeWitte, Matthew, and **Gomes, Rahul**. “Analyzing the Impact of Geospatial Derivatives on Domain Adaptation with CycleGAN.” *2024 IEEE International Conference on Electro Information Technology (eIT)*, pp. 710–715, 2024. <https://doi.org/10.1109/eIT60633.2024.10609908>
16. Tara, Vivek, Mitra, Dipankar, Muduganti, Aditi, Mali, Padmavathi, Maiti, Srabana, Dey, Shuvashis, and **Gomes, Rahul**. “On the Machine Learning-Based Multi-Class Classification of Microscopic Colitis.” *2024 IEEE International Conference on Electro Information Technology (eIT)*, pp. 38–43, 2024. <https://doi.org/10.1109/eIT60633.2024.10609954>
17. McKeown, Connor, Gillett, Philip, McCallum, Katherine, Meyer, Chloe, Mitchell, Nora, and **Gomes, Rahul**. “An Automated Deep Learning Approach for Analyzing Stomatal Morphometry of Poplar Trees.” *2024 IEEE International Conference on Electro Information Technology (eIT)*, pp. 1–6, 2024. <https://doi.org/10.1109/eIT60633.2024.10609904>
18. Konduru, Hemachandra, Callies, Benjamin, Mitra, Dipankar, Roy, Sayan, Dey, Shuvashis, and **Gomes, Rahul**. “On the Feasibility of a 3D-Printable Conductive Filament-Based Antenna Array for Conformal Applications.” *2024 IEEE International Symposium on Antennas and Propagation and INC/USNC-URSI Radio Science Meeting (AP-S/INC-USNC-URSI)*, pp. 423–424, 2024.
19. **Gomes, Rahul**, Pham, Tyler, He, Nichol, Kamrowski, Connor, and Wildenberg, Joseph. “Analysis of Swin-UNet Vision Transformer for Inferior Vena Cava Filter Segmentation from CT Scans.” *Artificial Intelligence in the Life Sciences*, 4:100084, 2023. <https://doi.org/10.1016/j.ailsci.2023.100084>
20. **Gomes, Rahul**, Denison Kroschel, Ashleigh, Day, Stephanie, and Jansen, Rick. “High Variation Across E. coli Hybrid Isolates Identified in Metabolism-Related Biological Pathways Co-Expressed with Virulent Genes.” *Gut Microbes*, 15(1):2228042, 2023. <https://doi.org/10.1080/19490976.2023.2228042>
21. Hasan, Munjur, Rahman, Md Saifur, Islam, Sabrina, Ahmed, Tanvir, Rifat, Nafiz, Ahsan, Mostofa, **Gomes, Rahul**, and Chowdhury, Md. “Vision Transformer-Based Classification for Lung and Colon Cancer Using Histopathology Images.” *2023 International Conference on Machine Learning and Applications (ICMLA)*, pp. 1300–1304, 2023. <https://doi.org/10.1109/ICMLA58977.2023.00196>
22. Ghosh, Abhimanyu, Kunkel, William, Varghese, Anthony, Ma, Ying, **Gomes, Rahul**, Bhattacharyya, Sudeep, Mohr, Molly, Doss, Isabella, and Hebert, Jordan. “Inter-Institutional Resource Sharing in Undergraduate HPC Education: Interviews with University Administrators.” *Proceedings of the 54th ACM Technical Symposium on Computer Science Education*, pp. 848–853, 2023. <https://doi.org/10.1145/3545945.3569784>
23. Rahman, Md Saifur, Rifat, Nafiz, Ahsan, Mostofa, Islam, Sabrina, Chowdhury, Md, and **Gomes, Rahul**. “Deep Learning Application for Detection of Malaria.” *2023 IEEE International Conference on Electro Information Technology (eIT)*, pp. 1–5, 2023. <https://doi.org/10.1109/eIT57321.2023.10187342>
24. Chowdhury, Md Minhaz, Rifat, Nafiz, Ahsan, Mostofa, Latif, Shadman, **Gomes, Rahul**, and Rahman, Md Saifur. “ChatGPT: A Threat Against the CIA Triad of Cyber Security.” *2023 IEEE International Conference on Electro Information Technology (eIT)*, pp. 1–6, 2023. <https://doi.org/10.1109/eIT57321.2023.10187355>

25. Rozario, Papia F., Ruehmann, Eleana, Pham, Tyler, Sun, Tianqi, Jensen, Jacob, Jia, Hengrui, Yu, Zhongyue, and **Gomes, Rahul**. “Deep Learning Patch-Based Approach for Hyperspectral Image Classification.” *2023 IEEE International Conference on Electro Information Technology (eIT)*, pp. 458–463, 2023. <https://doi.org/10.1109/eIT57321.2023.10187387>
26. Chowdhury, Minhaz, Rifat, Nafiz, Latif, Shadman, Ahsan, Mostofa, Rahman, Md Saifur, and **Gomes, Rahul**. “ChatGPT: The Curious Case of Attack Vectors’ Supply Chain Management Improvement.” *2023 IEEE International Conference on Electro Information Technology (eIT)*, pp. 499–504, 2023. <https://doi.org/10.1109/eIT57321.2023.10187385>
27. **Gomes, Rahul**, Kamrowski, Connor, Mohan, Pavithra Devy, Senor, Cameron, Langlois, Jordan, and Wildenberg, Joseph. “Application of Deep Learning to IVC Filter Detection from CT Scans.” *Diagnostics*, 12(10):2475, 2022. <https://doi.org/10.3390/diagnostics12102475>
28. **Gomes, Rahul**, Paul, Nijhum, He, Nichol, Huber, Aaron Francis, and Jansen, Rick J. “Application of Feature Selection and Deep Learning for Cancer Prediction Using DNA Methylation Markers.” *Genes*, 13(9):1557, 2022. <https://doi.org/10.3390/genes13091557>
29. Denton, Anne M., **Gomes, Rahul**, Schwartz, David M., and Franzen, David W. “Large-Window Curvature Computations for High-Resolution Digital Elevation Models.” *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–20, 2022. <https://doi.org/10.1109/TGRS.2022.3200354>
30. **Gomes, Rahul**, Kamrowski, Connor, Langlois, Jordan, Rozario, Papia, Dircks, Ian, Grottodden, Keegan, Martinez, Matthew, Tee, Wei Zhong, Sargeant, Kyle, LaFleur, Corbin, and Haley, Mitchell. “A Comprehensive Review of Machine Learning Used to Combat COVID-19.” *Diagnostics*, 12(8):1853, 2022. <https://doi.org/10.3390/diagnostics12081853>
31. Ahsan, Mostofa, Nygard, Kendall E., **Gomes, Rahul**, Chowdhury, Md Minhaz, Rifat, Nafiz, and Connolly, Jayden F. “Cybersecurity Threats and Their Mitigation Approaches Using Machine Learning—A Review.” *Journal of Cybersecurity and Privacy*, 2(3):527–555, 2022. <https://doi.org/10.3390/jcp2030027>
32. Rifat, Nafiz, Ahsan, Mostofa, Chowdhury, Md, and **Gomes, Rahul**. “BERT Against Social Engineering Attack: Phishing Text Detection.” *2022 IEEE International Conference on Electro Information Technology (eIT)*, pp. 1–6, 2022. <https://doi.org/10.1109/eIT53891.2022.9813922>
33. Ahsan, Mostofa, Rifat, Nafiz, Chowdhury, Md, and **Gomes, Rahul**. “Detecting Cyber Attacks: A Reinforcement Learning Based Intrusion Detection System.” *2022 IEEE International Conference on Electro Information Technology (eIT)*, pp. 461–466, 2022. <https://doi.org/10.1109/eIT53891.2022.9813892>
34. Ahsan, Mostofa, Rifat, Nafiz, Chowdhury, Md, and **Gomes, Rahul**. “Intrusion Detection for IoT Network Security with Deep Neural Network.” *2022 IEEE International Conference on Electro Information Technology (eIT)*, pp. 467–472, 2022. <https://doi.org/10.1109/eIT53891.2022.9814006>
35. Rifat, Nafiz, Ahsan, Mostofa, **Gomes, Rahul**, and Chowdhury, Md. “COVID-19 Sentiment Analysis Applying BERT.” *2022 IEEE International Conference on Electro Information Technology (eIT)*, pp. 417–422, 2022. <https://doi.org/10.1109/eIT53891.2022.9813777>
36. Ahsan, Mostofa, **Gomes, Rahul**, Chowdhury, Md Minhaz, and Nygard, Kendall E. “Enhancing Machine Learning Prediction in Cybersecurity Using Dynamic Feature Selector.” *Journal of Cybersecurity and Privacy*, 1(1):199–218, 2021. <https://doi.org/10.3390/jcp1010011>
37. Rozario, Papia F. and **Gomes, Rahul**. “Comparison of Data Mining Algorithms in Remote Sensing Using Lidar Data Fusion and Feature Selection.” *2021 IEEE International Conference on Electro/Information Technology (EIT)*, pp. 236–243, 2021. <https://doi.org/10.1109/EIT51626.2021.9491878>
38. **Gomes, Rahul**, Rozario, Papia F., and Adhikari, Nishan. “Deep Learning Optimization in Remote Sensing Image Segmentation Using Dilated Convolutions and ShuffleNet.” *2021 IEEE International Conference on Electro/Information Technology (EIT)*, pp. 244–249, 2021. <https://doi.org/10.1109/EIT51626.2021.9491910>
39. **Gomes, Rahul**, Denton, Anne, and Franzen, David. “Quantifying Efficiency of Sliding-Window Based Aggregation Technique by Using Predictive Modeling on Landform Attributes Derived from DEM and NDVI.” *ISPRS International Journal of Geo-Information*, 8(4):196, 2019. <https://doi.org/10.3390/ijgi8040196>

40. **Gomes, Rahul**, Denton, Anne, and Franzen, David. "Comparing Classification Accuracy of NDVI with DEM Derived Attributes Using Multi-Scalar Approach in Geographic Information Systems." *2019 IEEE International Conference on Electro Information Technology (EIT)*, pp. 249–254, 2019. <https://doi.org/10.1109/EIT.2019.8833766>
41. **Gomes, Rahul**, Denton, Anne, and Straub, Jeremy. "Comparative Study of Fitness Function in Genetic Algorithm for Optimal Site Allocation Using Lidar." *2019 IEEE International Conference on Electro Information Technology (EIT)*, pp. 255–260, 2019. <https://doi.org/10.1109/EIT.2019.8833664>
42. Ahsan, Mostofa, **Gomes, Rahul**, and Denton, Anne. "Application of a Convolutional Neural Network Using Transfer Learning for Tuberculosis Detection." *2019 IEEE International Conference on Electro Information Technology (EIT)*, pp. 427–433, 2019. <https://doi.org/10.1109/EIT.2019.8833768>
43. **Gomes, Rahul**, Ahsan, Mostofa, and Denton, Anne. "Random Forest Classifier in SDN Framework for User-Based Indoor Localization." *2018 IEEE International Conference on Electro/Information Technology (EIT)*, pp. 537–542, 2018. <https://doi.org/10.1109/EIT.2018.8500111>
44. Denton, Anne, **Gomes, Rahul**, and Franzen, David. "Scaling Up Window-Based Slope Computations for Geographic Information System." *2018 IEEE International Conference on Electro/Information Technology (EIT)*, pp. 554–559, 2018. <https://doi.org/10.1109/EIT.2018.8500288>
45. Ahsan, Mostofa, **Gomes, Rahul**, and Denton, Anne. "SMOTE Implementation on Phishing Data to Enhance Cybersecurity." *2018 IEEE International Conference on Electro/Information Technology (EIT)*, pp. 531–536, 2018. <https://doi.org/10.1109/EIT.2018.8500086>
46. Miryala, Goutham, **Gomes, Rahul**, and Karanam, Dayananda. "Comparative Analysis of Movie Recommendation System Using Collaborative Filtering in Spark Engine." *Journal of Global Research in Computer Science*, 8(10):10–14, 2017.
47. **Gomes, Rahul** and Straub, Jeremy. "Genetic Algorithm for Flood Detection and Evacuation Route Planning." *Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XXIII*, SPIE, vol. 10198, p. 384, 2017. <https://doi.org/10.1117/12.2266474>
48. Nelson, Ryan, Gabler, Andrew, Slusar, Skyler, Gordon, Aaron, McMillan, John, **Gomes, Rahul**, and Straub, Jeremy. "Additive Manufacturing (3D Printing) Material and Cost Reduction Algorithm Proof." *AIAA SPACE and Astronautics Forum and Exposition*, Paper 5225, 2017. <https://doi.org/10.2514/6.2017-5225>
49. Dayananda, Karanam Ravichandran, **Gomes, Rahul**, and Straub, Jeremy. "An Interconnected Architecture for an Emergency Medical Response Unmanned Aerial System." *2017 IEEE/AIAA 36th Digital Avionics Systems Conference (DASC)*, pp. 1–6, 2017. <https://doi.org/10.1109/DASC.2017.8102118>
50. **Gomes, Rahul**, Straub, Jeremy, Jones, Andrew, Morgan, John, Tipparach, Santipab, Sletten, Aaron, Kim, Keon Woo, Loegering, David, Feikema, Nick, Dayananda, Karanam, et al. "An Interconnected Network of UAS as a System-of-Systems." *2017 IEEE/AIAA 36th Digital Avionics Systems Conference (DASC)*, pp. 1–7, 2017.

SELECTED POSTERS AND ABSTRACTS

Underlined names indicate undergraduate student researchers.

1. Wildenberg, J., Kamrowski, C., Senor, C., Mohan, P., and **Gomes, R.** "Abstract No. 144: Automated IVC Filter Detection from Abdominopelvic CT Exams Using Deep Learning." *Journal of Vascular and Interventional Radiology*, 33(S), 2022. (Abstract presentation) <https://doi.org/10.1016/j.jvir.2022.03.225>
2. Adeleke, David, **Gomes, Rahul**, and Jansen, Rick. "Driver Gene Genomic Alterations Associated with Pancreatic Cancer Prognosis." *Society for Epidemiologic Research Annual Meeting*, Chicago, IL, 2022. (Poster presentation)
3. Paul, Nijhum, Jansen, Rick, **Gomes, Rahul**, He, Nichol, and Huber, Aaron. "A Scalable Deep Learning Framework for Breast Cancer Prediction Using DNA Methylation Data." *American Society of Human Genetics Annual Meeting*, 2021. (Poster presentation)
4. DeWitte, Matthew, Rozario, Papia F., Mohan, Pavithra Devy, and **Gomes, Rahul**. "Optimizing Deep Learning Architectures for Remote Sensing Image Analysis." *American Association of Geographers Annual Meeting*, 2021. (Poster presentation)

5. **Gomes, Rahul**, Durant, C., Chandra Deb, L., Ming, Y., Wang, Q., Zhao, J., Day, S., Bergholz, P., and Jansen, R. "Virulent Gene Expression Network Analysis and Visualization in E. coli." *80th Annual Meeting of the American Society for Microbiology*, 2020. (Poster presentation)
6. Denton, Anne, **Gomes, Rahul**, and Franzen, David. "Separating Landform from Noise in High-Resolution Digital Elevation Models through Scale-Adaptive Window-Based Regression." *International Conference on Spatial Statistics and Geostatistics*, New York, NY, 2019. (Abstract presentation)
7. Durant, C., **Gomes, Rahul**, Chandra Deb, L., Ming, Y., Wang, Q., Zhao, J., Day, S., Bergholz, P., and Jansen, R. "3D Visualization of Gene Expression Networks in E. coli." *6th AAPS-NDSU Research Symposium*, Fargo, ND, 2019. (Poster presentation)
8. **Gomes, Rahul** and Denton, Anne. "Incorporating Data Mining and Iterative Aggregation on Geospatial Datasets to Understand Soil Health in Depressions." *ND EPSCoR State Conference*, Fargo, ND, 2019. (Poster presentation)
9. **Gomes, Rahul** and Denton, Anne. "Taking Terrain Analysis to the Big Data Era for Understanding Soil Health in Depressions." *ND EPSCoR State Conference*, Grand Forks, ND, 2018. (Poster and abstract presentation)
10. **Gomes, Rahul** and Denton, Anne. "Application of Evolutionary Algorithms for Disaster Management and Response." *American Association of Geographers Annual Meeting*, New Orleans, LA, 2018. (Poster presentation)
11. **Gomes, Rahul**, Karanam, Dayananda, Straub, Jeremy, and Jones, Andrew. "Human Spaceflight Robotic Medical First Responder." *68th International Astronautical Congress*, Adelaide, Australia, 2017.
12. **Gomes, Rahul** and Straub, Jeremy. "Analysis of Multispectral Imaging Data and Genetic Algorithm-Based Approach Towards Disaster Management and Recovery." *North Dakota State University Graduate Research Symposium*, 2017. (Abstract presentation)

UNDERGRADUATE AND REGIONAL RESEARCH DISSEMINATION

Underlined names indicate undergraduate student researchers.

1. Morehead, Samuel, Tapia Martinez, Gabriel, Schuldes, Matt S., Blessing, Joshua M., Rich, Jennifer, and **Gomes, Rahul**. "Retrospective Institutional Comparison of Patients Undergoing On-Pump Versus Off-Pump Coronary Artery Bypass Grafting Using Machine Learning." *Mayo Clinic Health System Undergraduate Innovation Showcase*, Eau Claire, WI, November 2024. (Poster presentation)
2. Rolli, Alexandra, Mau, Brayden, **Gomes, Rahul**, Blair, David, and Harper, Sarah. "Harnessing the Web: Transforming Web-Scraped Data into Multi-Reading-Level Patient Education Resources for the MCHS Wisconsin Family Medicine Practice." *Mayo Clinic AI & I Palooza*, Rochester, MN, December 2024. (Poster presentation)
3. Rolli, Alexandra, Mau, Brayden, **Gomes, Rahul**, Blair, David, and Harper, Sarah. "Harnessing the Web: Transforming Web-Scraped Data into Multi-Reading-Level Patient Education Resources for the MCHS Wisconsin Family Medicine Practice." *Transforming Community and Rural Healthcare Symposium*, Rochester, MN, October 2024. (Poster presentation)
4. Rolli, Alexandra, Mau, Brayden, **Gomes, Rahul**, Blair, David, and Harper, Sarah. "Harnessing the Web: Transforming Web-Scraped Data into Multi-Reading-Level Patient Education Resources for the MCHS Wisconsin Family Medicine Practice." *2024 IEEE International Conference on Electro/Information Technology (EIT)*, Eau Claire, WI, 2024. (Poster presentation)
5. Rolli, Alexandra, Mau, Brayden, **Gomes, Rahul**, Garg, Sushil, and Harper, Sarah. "GI-What Now?: Creating an Open-Source Text-Extraction Tool for Reducing the Time to Diagnose Colonoscopy Results." *2024 IEEE International Conference on Electro/Information Technology (EIT)*, Eau Claire, WI, 2024. (Poster presentation)
6. Langlois, Jordan, Kamrowski, Connor, Martinez, Matthew, Dircks, Ian, Grottodden, Keegan, Haley, Mitchell, LaFleur, Corbin, Sargeant, Kyle, Tee, Wei Zhong, and **Gomes, Rahul**. "A Comprehensive Review of Artificial Intelligence Used to Combat COVID-19." *Celebration of Excellence in Research and Creative Activity (CERCA)*, University of Wisconsin-Eau Claire, 2022. (Poster presentation)

7. Kroschel, Ashleigh, Heinecke, Olivia, and **Gomes, Rahul**. “Comparison of Pathways and Detection Strategies for Pancreatic Ductal Adenocarcinoma (PDAC) Using Genetically Engineered Mouse Models (GEMM).” *Celebration of Excellence in Research and Creative Activity (CERCA)*, University of Wisconsin–Eau Claire, 2022. (Poster presentation)
8. Kamrowski, Connor, Mohan, Pavithra Devy, Senor, Cameron, and **Gomes, Rahul**. “Detection of Inferior Vena Cava Filters on CT Scans Using an Artificial Intelligence Algorithm.” *Celebration of Excellence in Research and Creative Activity (CERCA)*, University of Wisconsin–Eau Claire, 2022. (Poster presentation)
9. He, Nichol, Kamrowski, Connor, Varatharajan, Thulasi, Szymoniak, Amy, Lathiya, Maulik K., and **Gomes, Rahul**. “A Deep Learning Model for Pancreatic Ductal Adenocarcinoma Chemotherapy Outcome Prediction.” *Celebration of Excellence in Research and Creative Activity (CERCA)*, University of Wisconsin–Eau Claire, 2022. (Poster presentation)
10. Ma, Ying, **Gomes, Rahul**, Bhattacharyya, Sudeep, Mohr, Molly, Doss, Isabella, and Hebert, Jordan. “Exploring Policies to Promote High-Performance Computing in Post-Pandemic Undergraduate Education in Wisconsin.” *Celebration of Excellence in Research and Creative Activity (CERCA)*, University of Wisconsin–Eau Claire, 2022. (Poster presentation) <https://minds.wisconsin.edu/handle/1793/83867>
11. Huber, Aaron, Paul, Nijhum, **Gomes, Rahul**, and Jansen, Rick. “Identification of DNA Methylation Markers Using Feature Selection and Deep Learning.” *Celebration of Excellence in Research and Creative Activity (CERCA)*, University of Wisconsin–Eau Claire, 2021. (Poster presentation) <http://digital.library.wisc.edu/1793/82978>
12. Mohan, Pavithra Devy, DeWitte, Matthew, **Gomes, Rahul**, and Rozario, Papia F. “Optimizing Deep Learning Architecture for Remote Sensing Image Analysis.” *Celebration of Excellence in Research and Creative Activity (CERCA)*, University of Wisconsin–Eau Claire, 2021. (Poster presentation) <http://digital.library.wisc.edu/1793/83285>
13. Witham, Michael, Bender, Isaiah, and **Gomes, Rahul**. “Comparative Analysis of MariaDB’s Performance Efficiency as a Suitable Replacement for MySQL.” *Midwest Instruction and Computing Symposium (MICS)*, Fargo, ND, 2019. (Undergraduate conference paper)
14. Yi, Sumin and **Gomes, Rahul**. “Evaluating the Impact of Time Delays and Start Sequence for Effective Congestion Control Using TCP Reno, Westwood, and Vegas.” *Midwest Instruction and Computing Symposium (MICS)*, Fargo, ND, 2019. (Undergraduate conference paper)
15. Kamalanathan, Divyaa and **Gomes, Rahul**. “Comparing NoSQL and SQL Database Systems Based on Vulnerability to Injection and Adequacy of Countermeasures.” *Midwest Instruction and Computing Symposium (MICS)*, Fargo, ND, 2019. (Undergraduate conference paper)
16. Griswold, Gary, Carrato, Marcel Martinez, and **Gomes, Rahul**. “Quantitative Analysis to Verify Fairness of TCP CUBIC in NS-2.” *Midwest Instruction and Computing Symposium (MICS)*, Fargo, ND, 2019. (Undergraduate conference paper)
17. **Gomes, Rahul**, Ahsan, Mostofa, and Denton, Anne. “Fusion of SMOTE and Outlier Detection Techniques for Land-Cover Classification Using Support Vector Machines.” *Midwest Instruction and Computing Symposium (MICS)*, Duluth, MN, 2018. (Conference paper)

TEACHING EXPERIENCE

University of Wisconsin–Eau Claire

- CS 426 – Deep Learning 2021–Present
- CS 425 – Machine Learning 2024–Present
- CS 330 – Programming Languages 2022–Present
- CS 462 – Computer Networks 2023–2025
- CS 485 – Software Engineering II 2024

• CS 388 – Unix Programming	2023
• CS 396 – Junior Seminar	2021–2023
• CS 268 – Web Systems	2020–2024
• CS 376 – Cryptography and Network Security	2021
• CS 318 – Fundamentals of Web Page Design	2021–2022
• CS 146 – Big Picture in Computer Science	2021–2023
• CS 145 – Programming for New Programmers	2020
• CS 491 – Special Topics: Data Mining	2021

Curriculum and Course Development

- Contributed substantially to development of UW–Eau Claire’s B.A. and B.S. programs in Artificial Intelligence, launched in Fall 2025.
- Developed and revised curriculum in machine learning and deep learning, including CS 425 Machine Learning and CS 426 Deep Learning.
- Contributed to development of new artificial intelligence coursework, including foundations of AI, human–computer interaction, and natural language processing/generative AI.

Minot State University

• CSCI 111 – Introduction to Web Languages	2018, 2019
• CSCI 160 – Computer Science I	2019, 2020
• CSCI 340 – Computer Networks	2018, 2019
• CSCI 356 – Database Management I	2018, 2019
• CSCI 450 – Operating Systems	2020
• CSCI 456 – Database Management II	2019
• CSCI 458 – Network Security	2019, 2020

Other Teaching

- Dakota College at Bottineau: ASC 94 – Beginning Algebra. 2019
- North Dakota State University: CSCI 114 – Microcomputer Packages, Teaching Assistant. 2017

PROFESSIONAL AND EDITORIAL SERVICE

• General Chair , IEEE International Conference on Electro/Information Technology (EIT), UW–Eau Claire.	2024
• Publications Chair , IEEE International Conference on Electro/Information Technology.	2022–2023
• Session Chair , IEEE International Conference on Electro/Information Technology.	2021
• Academic Editor , PLOS ONE.	2022–Present
• Reviewer , National Science Foundation.	2023–Present
• Abstract Review Leader , National Conference on Undergraduate Research (NCUR), UW–Eau Claire.	2023
• Vice Chair , IEEE Region 4 Twin Cities Section.	2022
• Secretary , IEEE Region 4 Twin Cities Section.	2021, 2023

- **Faculty Advisor**, UW–Eau Claire Student Association for Computing Machinery. 2021–2024
- **Faculty Senator**, UW–Eau Claire Department of Computer Science. 2021–2022
- **Member**, UW–Eau Claire Senate Academic Policies Committee. 2021–2022
- **Member**, UW–Eau Claire College of Arts and Sciences Curriculum Committee. 2021–2022
- **Reviewer** for journals including IEEE Access; IEEE Transactions on Industrial Informatics; ACM Transactions on Privacy and Security; Scientific Reports; European Radiology; Sensors; Remote Sensing; Applied Sciences; ISPRS International Journal of Geo-Information; Future Internet; Machine Learning and Knowledge Extraction; and others.

SELECTED INVITED TALKS

- “Revolutionizing Intelligence: Paradigm Shifts in AI Technology.” *UW–Eau Claire Ask a Scientist Series*. 2024
- “Developing Usable Machine Learning Frameworks for Biomedical Applications.” *WiSys SPARK Symposium*. 2022
- “How Big Data Is Changing the Healthcare Industry.” *WiSys Innovation Brainstorm in Partnership with Mayo Clinic and UW–Eau Claire*. 2021
- “Preparing for Graduate School.” *Student Association for Computing Machinery, UW–Eau Claire*. 2020
- “Optimizing Image Analysis Techniques for Processing High-Resolution Spatial Big Data.” Invited presentations at Denison University, St. Mary’s College of California, UW–Oshkosh, and UW–Eau Claire. 2019
- “Analysis of Multispectral Imaging Data and Genetic Algorithm-Based Approaches Toward Disaster Management and Recovery.” Invited presentations at Minot State University and Minnesota State University, Mankato. 2018

SELECTED MEDIA COVERAGE AND INSTITUTIONAL PROFILES

- “Mentoring Record Brings National Award to UW–Eau Claire’s Dr. Rahul Gomes.” *University of Wisconsin–Eau Claire*. 2026
- “Faculty Feature: Dr. Rahul Gomes.” *University of Wisconsin–Eau Claire*. 2025
- “UWEC Will Offer AI Majors Starting Fall 2025.” *University of Wisconsin–Eau Claire*. 2025
- “Tech Industry Job Tremors and AI Boom Propel Changes at Wisconsin’s Colleges.” *PBS Wisconsin*. 2025
- “Grab Your Goggles: The Ask a Scientist Series.” *Volume One*. 2024
- “UWEC Students Collaborate with Mayo Clinic on AI Model for Medical Reports.” *Chippewa Valley Times*. 2024
- “UWEC Student Researchers Explore How AI Can Help Understand Health Care.” *Leader-Telegram*. 2024
- “UW–Eau Claire Receives Major NSF Grant for Summer Undergraduate Research.” *Universities of Wisconsin*. 2022
- “UW–Eau Claire: Building Bridges Between Health Care and Higher Education to Enhance Patient Care.” *Universities of Wisconsin*. 2021
- “UW–Eau Claire, HPE Collaboration a Powerful Partnership.” *Universities of Wisconsin*. 2021
- “Powerful New Supercomputer Will Open Doors for Research and Education at UW–Eau Claire.” *Milwaukee Journal Sentinel*. 2021